

Prehomework

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Question 1. Define what is meant by a function from the set X to the set Y .

Proof. A function $f : X \rightarrow Y$, defined from set X to set Y is a mapping where every element in X has a corresponding elements in Y . $\square$

Question 2. Define what it means for a function to be "onto" (also called "surjective"), and what it means for a function to be "one to one" (also called injective).

Proof. A surjective function is a function $f : X \rightarrow Y$ where $\forall y \in Y \exists x \in X$ s.t $f(x) = y$ or each element in Y has a corresponding element in X , however it is not needed that every element in X maps to an element in Y . An injective or one-to-one function is a function $f : X \rightarrow Y$ where $\forall x \in X \exists$ a unique $y \in Y$ s.t $f(x) = y$, meaning each element in X maps to a unique element in Y . $\square$

Question 3. Define a natural operation $Hom(X, Y) \times Hom(Y, Z) \rightarrow Hom(X, Z)$ (it should be composition), we indicate this by $(f, g) = g \circ f$. Show this is "associative", i.e. show that $h \circ (g \circ f) = (h \circ g) \circ f$. Explain why $Hom(X, X)$ is a "monoid" – ie this set comes with a natural associative operation. Explain also why it has an identity element.

Proof. The sets $Hom(X, X)$, $Hom(Y, Y)$, and $Hom(Z, Z)$ or any sets of functions where both the input and output sets are the same are monoids under the binary operation of composition. They have a natural associative operation.

Let $a, b, c \in Hom(X, X)$. By definition, $Hom(X, X)$ is a set of functions whose input set and output set are equal to X . $\forall x \in X$

$$(a \circ (b \circ c))(x) = a((b \circ c)(x)) = a(b(c(x))) = (a \circ b \circ c)(x)$$

Similarly,

$$((a \circ b) \circ c)(x) = (a \circ b)(c(x)) = a(b(c(x))) = (a \circ b \circ c)(x)$$

Because $\forall x \in X$, $(a \circ (b \circ c))(x) = (a \circ b \circ c)(x) = ((a \circ b) \circ c)(x)$, we have shown that the functions $(a \circ b) \circ c = a \circ (b \circ c)$. Thus the set $Hom(X, X)$ has a natural associativity operation under the binary operation of composition.

Furthermore, let $a \in Hom(X, X)$, where $a(x) = x$ for every $x \in X$. For every $b \in Hom(X, X)$, By definition, $(a \circ b)(x) = a(b(x)) = b(x)$ so $a \circ b = b$. Furthermore, by definition, $(b \circ a)(x) = b(a(x)) = b(x)$, so $b \circ a = b$. Since $a \circ b = b = b \circ a$ for every $b \in Hom(X, X)$, a must be the identity element in $Hom(X, X)$.

Thus because $Hom(X, X)$ has a natural associativity operation and a identity element, it is a monoid.

Let $f \in \text{Hom}(X, Y)$ and $g \in \text{Hom}(Y, Z)$. Because $\text{Hom}(X, Y) \times \text{Hom}(Y, Z) \rightarrow \text{Hom}(X, Z)$ it means $g \circ f = a \in \text{Hom}(X, Z)$. Finally, let $h \in \text{Hom}(Z, Z)$. By definition, $\text{Hom}(X, Y)$ is a set of functions whose input set is X and output set Y , $f : X \rightarrow Y$. Similarly by definition, $\text{Hom}(Y, Z)$ is a set of functions whose input set is Y and output set is Z , meaning $g : Y \rightarrow Z$. Finally, by definition $\text{Hom}(Z, Z)$ is a set of functions whose input and output sets are Z , thus $h : Z \rightarrow Z$.

For all $x \in X$, $(h \circ (g \circ f))(x) = h((g \circ f)(x)) = h(g(f(x))) = (h \circ g \circ f)(x)$ and $((h \circ g) \circ f)(x) = (h \circ g)(f(x)) = h(g(f(x))) = (h \circ g \circ f)(x)$.

Since $\forall x \in X$, $(h \circ (g \circ f))(x) = (h \circ g \circ f)(x) = ((h \circ g) \circ f)(x)$, $h \circ (g \circ f) = (h \circ g) \circ f$. Thus associativity is observed. $\square$

Question 4a. We start with $p : X \rightarrow Q$ and $F : X \rightarrow Z$. And the question is: Under what conditions does F "factor through p "? I.e. when does there exist a function $f : Q \rightarrow Z$ such that $F = f \circ p$.

$f : Q \rightarrow Z$ exists when $\forall a, b \in X$ if $p(a) = p(b)$ then $F(a) = F(b)$, because $p : X \rightarrow Q$ so if $p(a) = p(b)$ it means they map a and b to a 1 element in Q . Similarly if $F(a) = F(b)$, a and b are both mapped to 1 element in Z . If both of these events happen $\forall x \in X$, then just create a mapping f from the outputs of p to the outputs of F . However if there are elements of Q not in the image of p , map them to an arbitrary element in Z . Now for every $a, b \in X$ if $p(a) = p(b)$, $f(p(a)) = f(p(b))$ by the definition of a function and f exists s.t $F = f \circ p$. However if there exists $a, b \in X$, s.t $p(a) = p(b)$ however $F(a) \neq F(b)$, then no function f can be created because it would have to map one element in Q to 2 distinct elements in Z .

Question 4b. Show that that $f : Q \rightarrow Z$ exists iff $p(a) = p(b)$ implies $F(a) = F(b)$, for all a, b in X .

Proof. $\implies$ Suppose $f : Q \rightarrow Z$ exists. Select $a, b \in X$ s.t $p(a) = p(b)$. Since $F = f \circ p$, $F(a) = f(p(a)) = f(p(b)) = F(b)$. Thus, $p(a) = p(b) \implies F(a) = F(b)$.

$\impliedby$ Suppose $p(a) = p(b) \implies F(a) = F(b)$. Construct a function $f : Q \rightarrow Z$. Choose a $z \in Z$. Thus define f as the following:

$$f(q) = \begin{cases} F(p^{-1}(q)) & \text{if } q \in \text{Im}(p) \\ z & \text{if } q \in Q \setminus \text{Im}(p) \end{cases}$$

Since f maps an element in Q to a definite element in Z with no ambiguity, it is well defined. $\forall y \in X$ and by the definition of an image, $p(y) \in \text{Im}(p)$. So $f(p(y)) = F(p^{-1}(p(y))) = F(y)$. Thus $F(y) = (f \circ p)(y)$ and $F = f \circ p$ $\square$

Question 4c. Show that, assuming f exists, that it is unique iff p is surjective.

Counter Example: If the $|Z| = 1$, then there is a unique f even if p is not surjective. For example, if $X = \{1, 2, 3, 4\}$, $Q = \{a, b\}$, and $Z = \{\alpha\}$, then given the fact that f exists and $F = f \circ p$, in any case f would be a function which maps an element in Q to $\alpha \in Z$ even if p is not surjective. More specifically, $f(q) = \alpha$ and is unique while an example of a nonsurjective p is $p(x) = a$. Thus the condition that F factors over p is valid, f exists and is unique, however p is not surjective.

Proof. $\implies$ Suppose a function $f : Q \rightarrow Z$ s.t $F = f \circ p$ exists and is unique and $|Z| \neq 1$. Assume the contrary, that $p : X \rightarrow Q$ is not surjective. If p is not surjective, by the definition of a surjective function, it means $|Q| > |\text{Im}(p)|$ and $Q \setminus \text{Im}(p) \neq \emptyset$ meaning $\exists q \in Q \setminus \text{Im}(p)$.

1. If $|Z| = 0$, that means $|Q| = 0$ as well, because only an empty set can map to an empty set. However this contradicts the existence of q . Thus p must be surjective.

2. If $|Z| > 1$, then there will be more than 1 element in Z . Thus choose 2 unique elements $z_1, z_2 \in Z$. Create a function $f_1 : Q \rightarrow Z$ as the following:

$$f_1(q) = \begin{cases} F(p^{-1}(q)) & \text{if } q \in \text{Im}(p) \\ z_1 & \text{if } q \in Q \setminus \text{Im}(p) \end{cases}$$

Create a function $f_2 : Q \rightarrow Z$ as the following:

$$f_2(q) = \begin{cases} F(p^{-1}(q)) & \text{if } q \in \text{Im}(p) \\ z_2 & \text{if } q \in Q \setminus \text{Im}(p) \end{cases}$$

Since $z_1 \neq z_2$ and $\exists q \in Q \setminus \text{Im}(p)$, $f_1 \neq f_2$. Furthermore, $(f_1 \circ p)(x) = f_1(p(x))$. By the definition of a image, $f_1(p(x)) = F(p^{-1}(p(x))) = F(x)$. Thus $F = f_1 \circ p$. Similarly, $F = f_2 \circ p$. Both f_1 and f_2 fit the qualifications to be f , thus there are at least 2 choices for f , meaning f is not unique. This causes a contradiction, with our central assumption that f is unique. Thus by contradiction, p must be surjective.

$\Leftarrow$ Suppose a function $f : Q \rightarrow Z$ s.t $F = f \circ p$ exists and $|Z| \neq 1$. Assume p is surjective which means $Q = \text{Im}(p)$ and $Q \setminus \text{Im}(p) = \emptyset$. Which means $\forall q \in Q, q \in \text{Im}(p)$. Using the proof above, $f : Q \rightarrow Z$ can be defined as the following, for any $z \in Z$.

$$f(q) = \begin{cases} F(p^{-1}(q)) & \text{if } q \in \text{Im}(p) \\ z & \text{if } q \in Q \setminus \text{Im}(p) \end{cases}$$

However the second case can be disregarded since $Q \setminus \text{Im}(p) = \emptyset$ and there can be no $q \in \emptyset$. So the second condition is always false, the choice of z doesn't matter and $f(q) = F(p^{-1}(q))$. Thus there is no ambiguity in the definition of f , thus f is unique.

□